

| Question | Answer  | Mark | Mark scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Additional guidance                                                                                                                                                                        |
|----------|---------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21       | -6, 3.5 | P1   | <p>for process to find gradient of <math>PR</math> or <math>QR</math>,<br/> eg <math>\frac{k+3-5}{k-4} \left( = \frac{k-2}{k+4} \right)</math> <b>or</b> <math>\frac{k+3-6}{k-6} \left( = \frac{k+9}{k-6} \right)</math> <b>or</b><br/> <math>\frac{5-(k+3)}{-4-k} \left( = \frac{2-k}{-4-k} \right)</math> <b>or</b> <math>\frac{-6-(k+3)}{6-k} \left( = \frac{-9-k}{6-k} \right)</math><br/> <b>OR</b><br/> for start of process to use Pythagoras,<br/> eg <math>PR^2 = (k-4)^2 + (k+3-5)^2</math> <b>or</b> <math>RP^2 = (-4-k)^2 + (-5-(k+3))^2</math><br/> <b>or</b> <math>QR^2 = (k-6)^2 + (k+3-6)^2</math> <b>or</b> <math>RQ^2 = (6-k)^2 + (-6-(k+3))^2</math></p> | <p>Condone missing bracket for first P1 only</p> <p>Stating eg Grad <math>PR = \frac{k-4}{k+3-5}</math></p> <p>Or grad <math>QR = \frac{k-6}{k+3-6}</math> scores P0<br/> but check SC</p> |
|          |         | P1   | <p>for forming a correct equation,<br/> eg <math>\frac{k+3-5}{k-4} \times \frac{k+3-6}{k-6} = -1</math> <b>or</b> <math>\frac{k-2}{k+4} \times \frac{k+9}{k-6} = -1</math> <b>or</b><br/> <math>\frac{5-(k+3)}{-4-k} \times \frac{-6-(k+3)}{6-k} = -1</math> <b>or</b> <math>\frac{2-k}{-4-k} \times \frac{-9-k}{6-k} = -1</math> oe<br/> <b>OR</b><br/> <math>(k-4)^2 + (k+3-5)^2 + (k-6)^2 + (k+3-6)^2 = (-6-5)^2 + (6-4)^2</math><br/> <b>or</b> <math>(k+4)^2 + (k-2)^2 + (k-6)^2 + (k+9)^2 = (-11)^2 + 10^2</math></p>                                                                                                                                                 |                                                                                                                                                                                            |
|          |         | P1   | <p>for writing in the form <math>ak^2 + bk + c (= 0)</math>,<br/> eg <math>2k^2 + 5k - 42 (= 0)</math> <b>or</b> <math>4k^2 + 10k - 84 (= 0)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                            |
|          |         | P1   | <p>(dep P3) for factorising, eg <math>(k+6)(2k-7) (= 0)</math> <b>or</b> <math>(2k+12)(2k-7)</math> <b>or</b><br/> <math>(k+6)(4k-14)</math><br/> <b>OR</b> use of formula, eg <math>\frac{-5 \pm \sqrt{5^2 - 4 \times 2 \times -42}}{2 \times 2}</math> <b>or</b> <math>\frac{-10 \pm \sqrt{10^2 - 4 \times 4 \times -84}}{2 \times 4}</math></p>                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                            |
|          |         | A1   | <p>cao<br/> SCB4 for correct answer coming from consistent use of reciprocals of gradients</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                            |

## **Documented scheme clarification**

1MA1\_NOV\_2024\_2H\_Q21

## **Original source**

scheme page 20, SHA256

9d7ee2fd705b607c70423ec1b7fedc4940917c4295e50bc0b3d3cb95cc1b1fa8

## **Original wording**

$$RP^2 = (-4-k)^2 + (-5-(k+3))^2$$

## **Correction / interpretation**

$$RP^2 = (-4-k)^2 + (5-(k+3))^2$$

## **Reason**

P has y-coordinate 5. Use 5 in the squared-distance alternative; the official source pixels are retained.

## **Sol authority**

85f03d0210de426aaffb8f1abc0574ef68109c99c10907277b51710829ed00f0